

arr[9]

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
33	55	88	77	44	11	66	22	99

- 1) Get the key from the user.
- 2) Start the traversal from the first element.[0th index].
- 3) **Compare** the key with each element from the array.
- 4) If the key matches the element, return the corresponding index.
Else, continue the search till the last index.

```

For(i = 0; i < 9; i++)
{
    If (key == arr[i])
        Return i;
}

```

Key = 77
Key found at index 3.
Comparisons = 4.

Average Case Time Complexity

$O(n/2) \rightarrow O(n) \rightarrow \Theta(n)$
 $1/2$ is discarded.

Key = 33
Key found at index 0.
Comparisons = 1.

Best Case Time Complexity

Order of $O(1) \rightarrow \Omega(1)$
 $O(1)$

Key = 99
Key found at index 8
Comparisons = 9.

Worst Case Time Complexity

$\checkmark O(n)$
 $n \rightarrow$ number of elements $\rightarrow O(n)$

Key = 10
Key Not Found.
Comparisons = 9.

Worst Case Time complexity

$O(n) \rightarrow O(n)$